



**Maynooth
University**

National University
of Ireland Maynooth

**SEMESTER 1
2022-2023**

**CS264FZ
Software Design**

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Time allowed: 2 hours

Answer *four* questions

All questions carry equal marks

Instructions

	Yes	No
Log Books Allowed	☐	☒
Formula Tables Allowed	☐	☒
Other Allowed (<i>enter details</i>)	☐	☒

General (*enter detail*)

QUESTION 1

- (a) What are the three roles in the chain-of-responsibility pattern? Please describe each role's tasks. (5 marks)
- (b) When should you consider implementing the visitor pattern? (5 marks)
- (c) What are the key benefits of the observer pattern? (5 marks)
- (d) What is the difference between lazy initialization and eager initialization? (5 marks)
- (e) Why does the decorator pattern prefer composition over class inheritance in realizing dynamic addition? (5 marks)

QUESTION 2

- (a) Can extrinsic data be shared in the flyweight pattern? How do clients handle extrinsic data? (4 marks)
- (b) In the mediator pattern, suppose that you are reducing the number of interconnections among various objects. What key benefits will you achieve from the reduction? (4 marks)
- (c) You could use simple subclassing instead of the bridge pattern. Is this correct? Please give your reasons. (5 marks)
- (d) Please introduce the participants of the iterator pattern. (6 marks)
- (e) What is the difference between the facade pattern and the adapter pattern? (6 marks)

QUESTION 3

- (a) Use the interpreter pattern to simulate the addition, subtraction, and multiplication of complex numbers.

Requirements:

- (i) Write a ComplexNum class that has two attributes to represent the real and imaginary parts of a complex number. Write a Calculate class that has two attributes leftNum and rightNum. Write an AbstractExpression interface with an interpret(Calculate calculate) method. (5 mark)
- (ii) Write an AddExpression class that implements the AbstractExpression interface. The interpret() method is implemented as the addition of two complex numbers. (5 mark)
- (iii) Write a SubExpression class that implements the AbstractExpression interface. The interpret() method is implemented as the subtraction of two complex numbers. (5 mark)
- (iv) Write a MultiplyExpression class that implements the AbstractExpression interface. The interpret() method is implemented as the multiplication of two complex numbers. (5 mark)
- (v) Write a Test class to test the above complex number operations. (5 mark)

QUESTION 4

- (a) Please use the abstract factory pattern to simulate the creation of Huawei and Xiaomi mobile phones and computers by factory classes.
- (i) Create a Phone interface with a phoneBrand() method. (3 mark)
 - (ii) Create two classes with the names of HuaweiPhone and XiaomiPhone that implement the Phone interface. (6 mark)
 - (iii) Create a Computer interface with a computerBrand() method. (4 mark)
 - (iv) Create a Factory interface with a factoryPhone() and a factoryComputer() methods. (4 mark)
 - (v) Create two classes with the names of HuaweiFactory and XiaomiFactory that implement the Factory interface. The factoryPhone() and factoryComputer() methods are implemented as creating a HuaweiPhone and a XiaomiPhone object, respectively. (8 mark)